2.10.43

EE25BTECH11014 - Bhoomika Lokesh

Question:

Let $\mathbf{a} = \mathbf{i} - \mathbf{k}, \mathbf{b} = x\mathbf{i} + \mathbf{j} + (1-x)\mathbf{k}$ and $\mathbf{C} = y\mathbf{i} + x\mathbf{j} + (1+x-y)\mathbf{k}$.Then $[\mathbf{a}\ \mathbf{b}\ \mathbf{c}]$ depends on

1) only x
2) only y
3) Neither x nor y
4) both x and y

Solution:

$$[\mathbf{a}\ \mathbf{b}\ \mathbf{c}] = \begin{vmatrix} 1 & 0 & -1 \\ x & 1 & 1-x \\ y & x & 1+x-y \end{vmatrix} \tag{1}$$

Converting the above matrix into Row Echelon form,

$$R_2 \longrightarrow R_2 - xR_1 \implies \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ y & x & 1+x-y \end{pmatrix} \tag{2}$$

$$R_3 \longrightarrow R_3 - yR_1 \implies \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & x & 1+x \end{pmatrix} \tag{3}$$

$$R_3 \longrightarrow R_3 - xR_2 \implies \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \tag{4}$$

$$[\mathbf{a}\ \mathbf{b}\ \mathbf{c}] = (1)\begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} - (0)\begin{vmatrix} 0 & 1 \\ 0 & 1 \end{vmatrix} + (-1)\begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix} = 1 \tag{5}$$

$$\implies [\mathbf{a}\ \mathbf{b}\ \mathbf{c}] = 1 \tag{6}$$

$\therefore$ $[\mathbf{a}\ \mathbf{b}\ \mathbf{c}]$ is neither dependent on x and y.